Part 2 Cloud native development

Part 1 defined the main features of cloud native applications, and you got a first taste of the journey from code to deployment. Part 2 will introduce you to the main practices and patterns for building production-ready cloud native applications using Spring Boot and Kubernetes.

Chapter 3 covers the fundamentals of starting a new cloud native project, including strategies for organizing the codebase, managing dependencies, and defining the commit stage of a deployment pipeline. You'll learn how to implement and test a REST API using Spring MVC and Spring Boot Test. Chapter 4 discusses the importance of externalized configuration and covers some options available for Spring Boot applications, including property files, environment variables, and configuration services with Spring Cloud Config. Chapter 5 presents the main aspects of data services in the cloud and shows how to add data persistence to a Spring Boot application using Spring Data JDBC. You'll learn production options for managing data using Flyway and testing strategies using Testcontainers. Chapter 6 is about containers; you'll learn more about Docker and how to package Spring Boot applications as container images using Dockerfiles and Cloud Native Buildpacks. Finally, chapter 7 discusses Kubernetes and covers service discovery, load balancing, scalability, and local development workflows. You'll also learn more about how to deploy Spring Boot applications to a Kubernetes cluster.